Brutus_Sherlock

18 April 2026 20:06

NOTE: This HTB Sherlock has been done independently. The resources are given in the HTB Sherlock page.

1. Upon listing the files in the `Brutus` directory, we can see three files.

```
kali@kali: ~/HTB/Brutus
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/HTB/Brutus]
└─$ ls
auth.log  utmp.py  wtmp

┌──(kali㉿kali)-[~/HTB/Brutus]
└─$ cat auth.log
Mar  6 06:18:01 ip-172-31-35-28 CRON[1119]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:18:01 ip-172-31-35-28 CRON[1118]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:18:01 ip-172-31-35-28 CRON[1117]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:18:01 ip-172-31-35-28 CRON[1118]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:18:01 ip-172-31-35-28 CRON[1119]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:18:01 ip-172-31-35-28 CRON[1117]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:19:01 ip-172-31-35-28 CRON[1366]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:19:01 ip-172-31-35-28 CRON[1367]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:19:01 ip-172-31-35-28 CRON[1366]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:19:01 ip-172-31-35-28 CRON[1367]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:19:52 ip-172-31-35-28 sshd[1465]: AuthorizedKeysCommand /usr/share/ec2-instance-connect/eic_run_authorized_keys ro
A failed, status 22
Mar  6 06:19:54 ip-172-31-35-28 sshd[1465]: Accepted password for root from [redacted] 9 port 42825 ssh2
Mar  6 06:19:54 ip-172-31-35-28 sshd[1465]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=0)
Mar  6 06:19:54 ip-172-31-35-28 systemd-logind[411]: New session 6 of user root.
Mar  6 06:19:54 ip-172-31-35-28 systemd: pam_unix(systemd-user:session): session opened for user root(uid=0) by (uid=0)
Mar  6 06:20:01 ip-172-31-35-28 CRON[1599]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:20:01 ip-172-31-35-28 CRON[1600]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:20:01 ip-172-31-35-28 CRON[1599]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:20:01 ip-172-31-35-28 CRON[1600]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:21:01 ip-172-31-35-28 CRON[1628]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:21:01 ip-172-31-35-28 CRON[1629]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:21:01 ip-172-31-35-28 CRON[1629]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:21:01 ip-172-31-35-28 CRON[1628]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:22:01 ip-172-31-35-28 CRON[1650]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:22:01 ip-172-31-35-28 CRON[1649]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:22:01 ip-172-31-35-28 CRON[1649]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:22:01 ip-172-31-35-28 CRON[1650]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:23:01 ip-172-31-35-28 CRON[2183]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:23:01 ip-172-31-35-28 CRON[2184]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:23:01 ip-172-31-35-28 CRON[2183]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:23:01 ip-172-31-35-28 CRON[2184]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:24:01 ip-172-31-35-28 CRON[2199]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:24:01 ip-172-31-35-28 CRON[2198]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:24:01 ip-172-31-35-28 CRON[2199]: pam_unix(cron:session): session closed for user confluence
```

2. For Task 1, we can see an external system with IP trying to gain a foothold on the `Confluence` server via SSH service brute-forcing.

```
Mar  6 06:31:31 ip-172-31-35-28 sshd[2325]: Invalid user admin from [redacted] 68 port 46380
Mar  6 06:31:31 ip-172-31-35-28 sshd[2325]: Received disconnect from [redacted].68 port 46380:11: Bye Bye [preauth]
Mar  6 06:31:31 ip-172-31-35-28 sshd[2325]: Disconnected from invalid user admin [redacted].68 port 46380 [preauth]
Mar  6 06:31:31 ip-172-31-35-28 sshd[620]: error: beginning MaxStartups throttling
Mar  6 06:31:31 ip-172-31-35-28 sshd[620]: drop connection #10 from [[redacted] 68]:46482 on [172.31.35.28]:22 past MaxStartup
Mar  6 06:31:31 ip-172-31-35-28 sshd[2327]: Invalid user admin from [redacted].68 port 46392
Mar  6 06:31:31 ip-172-31-35-28 sshd[2327]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2327]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2332]: Invalid user admin from [redacted].68 port 46444
Mar  6 06:31:31 ip-172-31-35-28 sshd[2331]: Invalid user admin from [redacted] 68 port 46436
Mar  6 06:31:31 ip-172-31-35-28 sshd[2332]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2332]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2331]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2331]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2330]: Invalid user admin from [redacted] 68 port 46422
Mar  6 06:31:31 ip-172-31-35-28 sshd[2337]: Invalid user admin from [redacted].68 port 46498
Mar  6 06:31:31 ip-172-31-35-28 sshd[2328]: Invalid user admin from [redacted].68 port 46390
Mar  6 06:31:31 ip-172-31-35-28 sshd[2335]: Invalid user admin from [redacted].68 port 46460
Mar  6 06:31:31 ip-172-31-35-28 sshd[2337]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2337]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2335]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2335]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2334]: Invalid user admin from [redacted] 68 port 46454
Mar  6 06:31:31 ip-172-31-35-28 sshd[2329]: Invalid user admin from [redacted].68 port 46414
Mar  6 06:31:31 ip-172-31-35-28 sshd[2338]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2330]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2334]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2334]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2336]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2330]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2328]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2328]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser
Mar  6 06:31:31 ip-172-31-35-28 sshd[2333]: Invalid user admin from [redacted].68 port 46452
```

3. For Task 2, we can see the attacker has successfully gain access to `root` account.

```
Mar  6 06:32:39 ip-172-31-35-28 sshd[620]: exited MaxStartups throttling after 00:01:08, 21 connections dropped
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: Accepted password for root from [redacted].68 port 53184 ssh2
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=
Mar  6 06:32:44 ip-172-31-35-28 systemd-logind[411]: New session 37 of user root.
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/gshadow: name=cyberjunkie
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: new group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 useradd[2592]: new user: name=cyberjunkie, UID=1002, GID=1002, home=/home/cyber
Mar  6 06:34:26 ip-172-31-35-28 passwd[2603]: pam_unix(passwd:chauthtok): password changed for cyberjunkie
Mar  6 06:34:31 ip-172-31-35-28 chfn[2605]: changed user 'cyberjunkie' information
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session closed for user root
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to group 'sudo'
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to shadow group 'sudo'
```

4. For Task 3, we need to find the UTC timestamp when the attacker login's manually to the server. As from first image, the `wtmp` file is a binary log file in Linux systems that records the data of all user logins, logouts, system boots and shutdowns.

Since `wtmp` file is binary file, viewing the file will reveal binary, non-readable data. To make it readable the given `utmp.py` script converts the binary non-readable to ASCII-type readable.

```
kali@kali: ~/HTB/Brutus
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/HTB/Brutus]
└─$ python utmp.py wtmp
"type"  "pid"   "line"  "id"    "user"  "host"  "term"  "exit"  "session"       "sec"   "usec"  "addr"
"BOOT_TIME"     "0"     "~"     "~~"    "reboot"        "6.2.0-1017-aws"        "0"     "0"     "0"     "2024/01/25 06:12:17"   "804944"        "0.0.0.0"
"INIT"  "601"   "ttyS0" "tyS0"  ""      ""      "0"     "0"     "601"   "2024/01/25 06:12:31"   "72401" "0.0.0.0"
"LOGIN" "601"   "ttyS0" "tyS0"  "LOGIN" ""      "0"     "0"     "601"   "2024/01/25 06:12:31"   "72401" "0.0.0.0"
"INIT"  "618"   "tty1"  "tty1"  ""      ""      "0"     "0"     "618"   "2024/01/25 06:12:31"   "80342" "0.0.0.0"
"LOGIN" "618"   "tty1"  "tty1"  "LOGIN" ""      "0"     "0"     "618"   "2024/01/25 06:12:31"   "80342" "0.0.0.0"
"RUN_LVL"       "53"    "~"     "~~"    "runlevel"      "6.2.0-1017-aws"        "0"     "0"     "0"     "2024/01/25 06:12:33"   "792454"        "0.0.0.0"
"USER"  "1284"  "pts/0" "ts/0"  "ubuntu"        "[redacted].9"  "0"     "0"     "0"     "2024/01/25 06:13:58"   "354674"        "[redacted].9"
"DEAD"  "1284"  "pts/0" ""      ""      ""      "0"     "0"     "0"     "2024/01/25 06:15:12"   "956114"        "0.0.0.0"
"USER"  "1483"  "pts/0" "ts/0"  "root"  "[redacted].9"  "0"     "0"     "0"     "2024/01/25 06:15:40"   "806926"        "[redacted]9"
"DEAD"  "1404"  "pts/0" ""      ""      ""      "0"     "0"     "0"     "2024/01/25 07:34:34"   "949753"        "0.0.0.0"
"USER"  "836798"        "pts/0" "ts/0"  "root"  "[redacted].9"  "0"     "0"     "0"     "2024/02/11 05:33:49"   "408334"        "[redacted]9"
"INIT"  "838568"        "ttyS0" "tyS0"  ""      ""      "0"     "0"     "838568"        "2024/02/11 05:39:02"   "172417"        "0.0.0.0"
"LOGIN" "838568"        "ttyS0" "tyS0"  "LOGIN" ""      "0"     "0"     "838568"        "2024/02/11 05:39:02"   "172417"        "0.0.0.0"
"USER"  "838962"        "pts/1" "ts/1"  "root"  "[redacted].9"  "0"     "0"     "0"     "2024/02/11 05:41:11"   "700107"        "[redacted]9"
"DEAD"  "838896"        "pts/1" ""      ""      ""      "0"     "0"     "0"     "2024/02/11 05:41:46"   "272984"        "0.0.0.0"
"USER"  "842171"        "pts/1" "ts/1"  "root"  "[redacted].9"  "0"     "0"     "0"     "2024/02/11 05:54:27"   "775434"        "[redacted]9"
"DEAD"  "842073"        "pts/1" ""      ""      ""      "0"     "0"     "0"     "2024/02/11 06:08:04"   "769514"        "0.0.0.0"
"DEAD"  "836694"        "pts/0" ""      ""      ""      "0"     "0"     "0"     "2024/02/11 06:08:04"   "769963"        "0.0.0.0"
"RUN_LVL"       "0"     "~"     "~~"    "shutdown"      "6.2.0-1017-aws"        "0"     "0"     "0"     "2024/02/11 06:09:18"   "731"   "0.0.0.0"
"BOOT_TIME"     "0"     "~"     "~~"    "reboot"        "6.2.0-1018-aws"        "0"     "0"     "0"     "2024/03/06 01:17:15"   "744575"        "0.0.0.0"
"INIT"  "464"   "ttyS0" "tyS0"  ""      ""      "0"     "0"     "464"   "2024/03/06 01:17:27"   "354378"        "0.0.0.0"
"LOGIN" "464"   "ttyS0" "tyS0"  "LOGIN" ""      "0"     "0"     "464"   "2024/03/06 01:17:27"   "354378"        "0.0.0.0"
"INIT"  "505"   "tty1"  "tty1"  ""      ""      "0"     "0"     "505"   "2024/03/06 01:17:27"   "469940"        "0.0.0.0"
"LOGIN" "505"   "tty1"  "tty1"  "LOGIN" ""      "0"     "0"     "505"   "2024/03/06 01:17:27"   "469940"        "0.0.0.0"
"RUN_LVL"       "53"    "~"     "~~"    "runlevel"      "6.2.0-1018-aws"        "0"     "0"     "0"     "2024/03/06 01:17:29"   "538024"        "0.0.0.0"
"USER"  "1583"  "pts/0" "ts/0"  "root"  "[redacted].9"  "0"     "0"     "0"     "2024/03/06 01:19:55"   "151913"        "[redacted].9"
"USER"  "2549"  "pts/1" "ts/1"  "root"  "[redacted]68"  "0"     "0"     "0"     "2024/03/06 01:32:45"   "387923"        "[redacted]68"
"DEAD"  "2491"  "pts/1" ""      ""      ""      "0"     "0"     "0"     "2024/03/06 01:37:24"   "590579"        "0.0.0.0"
"USER"  "2667"  "pts/1" "ts/1"  "cyberjunkie"  "[redacted]68"  "0"     "0"     "0"     "2024/03/06 01:37:35"   "475575"        "[redacted].68"
```

From the above image, it is clear that the attacker logged on to server.
NOTE: There is a bug in the python script that prints the hour as '01' instead of '06'.

5. For Task 4, the SSH session number assigned to the attacker is present in the `auth.log` file. From the image, we can see that session number 37 is assigned. This shows that the attacker has successfully gained access to the `root` account through SSH brute-forcing,

```
Mar  6 06:32:39 ip-172-31-35-28 sshd[620]: exited MaxStartups throttling after 00:01:08, 21 connections dropped
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: Accepted password for root from [redacted].68 port 53184 ssh2
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=
Mar  6 06:32:44 ip-172-31-35-28 systemd-logind[411]: New session 37 of user root.
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/gshadow: name=cyberjunkie
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: new group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 useradd[2592]: new user: name=cyberjunkie, UID=1002, GID=1002, home=/home/cyber
Mar  6 06:34:26 ip-172-31-35-28 passwd[2603]: pam_unix(passwd:chauthtok): password changed for cyberjunkie
Mar  6 06:34:31 ip-172-31-35-28 chfn[2605]: changed user 'cyberjunkie' information
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session closed for user root
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to group 'sudo'
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to shadow group 'sudo'
```

6. For Task 5, the attacker created a group user to maintain foothold. This can be found in `auth.log`.

```
Mar  6 06:32:39 ip-172-31-35-28 sshd[620]: exited MaxStartups throttling after 00:01:08, 21 connections dropped
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: Accepted password for root from [redacted].68 port 53184 ssh2
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=
Mar  6 06:32:44 ip-172-31-35-28 systemd-logind[411]: New session 37 of user root.
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/gshadow: name=cyberjunkie
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: new group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 useradd[2592]: new user: name=cyberjunkie, UID=1002, GID=1002, home=/home/cyber
Mar  6 06:34:26 ip-172-31-35-28 passwd[2603]: pam_unix(passwd:chauthtok): password changed for cyberjunkie
Mar  6 06:34:31 ip-172-31-35-28 chfn[2605]: changed user 'cyberjunkie' information
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session opened for user confluence(uid=998)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session closed for user root
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to group 'sudo'
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to shadow group 'sudo'
```

7. For Task 6, the MITRE ATT&CK sub-technique used for persistence by creating a new account can be found in the official MITRE ATT&CK webpage. Persistence is given as id `TA0003`. Under which creating a local account is given `T1136.001` id.

MITRE | ATT&CK

Matrices · Tactics · Techniques · Defenses · CTI · Resources · Benefactors · Contribute · Blog

TACTICS

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Defense Evasion
- Credential Access
- Discovery
- Lateral Movement
- Collection
- Command and Control
- Exfiltration
- Impact

Mobile

ICS

T1554	Compromise Host Software Binary	Adversaries may modify host software binaries to establish persistent access to systems. Software binaries/executables provide a wide range of system commands or services, programs, and libraries. Common software binaries are SSH clients, FTP clients, email clients, web browsers, and many other user or server applications.
T1136	Create Account	Adversaries may create an account to maintain access to victim systems. With a sufficient level of access, creating such accounts may be used to establish secondary credentialed access that do not require persistent remote access tools to be deployed on the system.
.001	Local Account	Adversaries may create a local account to maintain access to victim systems. Local accounts are those configured by an organization for use by users, remote support, services, or for administration on a single system or service.
.002	Domain Account	Adversaries may create a domain account to maintain access to victim systems. Domain accounts are those managed by Active Directory Domain Services where access and permissions are configured across systems and services that are part of that domain. Domain accounts can cover user, administrator, and service accounts. With a sufficient level of access, the `net user /add /domain` command can be used to create a domain account.
.003	Cloud Account	Adversaries may create a cloud account to maintain access to victim systems. With a sufficient level of access, such accounts may be used to establish secondary credentialed access that does not require persistent remote access tools to be deployed on the system.
T1543	Create or Modify	Adversaries may create or modify system-level processes to repeatedly execute malicious payloads

8. For Task 7, we are asked to find the attacker's first SSH session end which can be found in `auth.log`.

```
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Received disconnect from [redacted]68 port 53184:11: disconnected by user
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Disconnected from user root [redacted].68 port 53184
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session closed for user root
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Session 37 logged out. Waiting for processes to exit.
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Removed session 37.
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: Accepted password for cyberjunkie from [redacted].68 port 43260 ssh2
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: pam_unix(sshd:session): session opened for user cyberjunkie(uid=1002) by (ui
Mar  6 06:37:34 ip-172-31-35-28 systemd-logind[411]: New session 49 of user cyberjunkie.
Mar  6 06:37:34 ip-172-31-35-28 systemd: pam_unix(systemd-user:session): session opened for user cyberjunkie(uid=1002) b
Mar  6 06:37:57 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cat
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=100
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:01 ip-172-31-35-28 CRON[2765]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:39:01 ip-172-31-35-28 CRON[2764]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:39:01 ip-172-31-35-28 CRON[2765]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:01 ip-172-31-35-28 CRON[2764]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:38 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cur
/main/linper.sh
Mar  6 06:39:38 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=100
Mar  6 06:39:39 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:40:01 ip-172-31-35-28 CRON[2783]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:40:01 ip-172-31-35-28 CRON[2784]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
```

9. For Task 8, the attacker has downloaded a script after obtaining access to the `root` account. From the image, the attacker has downloaded `linper`, which is a script that allows to enumerate programs that can execute a reverse shell to persist the shell session even after the target system reboots.

```
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Received disconnect from [redacted]68 port 53184:11: disconnected by user
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Disconnected from user root [redacted].68 port 53184
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session closed for user root
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Session 37 logged out. Waiting for processes to exit.
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Removed session 37.
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: Accepted password for cyberjunkie from [redacted].68 port 43260 ssh2
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: pam_unix(sshd:session): session opened for user cyberjunkie(uid=1002) by (ui
Mar  6 06:37:34 ip-172-31-35-28 systemd-logind[411]: New session 49 of user cyberjunkie.
Mar  6 06:37:34 ip-172-31-35-28 systemd: pam_unix(systemd-user:session): session opened for user cyberjunkie(uid=1002) b
Mar  6 06:37:57 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cat
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=100
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session closed for user confluence
```

```
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Received disconnect from [redacted]68 port 53184:11: disconnected by user
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Disconnected from user root [redacted].68 port 53184
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session closed for user root
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Session 37 logged out. Waiting for processes to exit.
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Removed session 37.
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: Accepted password for cyberjunkie from [redacted].68 port 43260 ssh2
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: pam_unix(sshd:session): session opened for user cyberjunkie(uid=1002) by (ui
Mar  6 06:37:34 ip-172-31-35-28 systemd-logind[411]: New session 49 of user cyberjunkie.
Mar  6 06:37:34 ip-172-31-35-28 systemd: pam_unix(systemd-user:session): session opened for user cyberjunkie(uid=1002) b
Mar  6 06:37:57 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cat
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=100
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:01 ip-172-31-35-28 CRON[2765]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:39:01 ip-172-31-35-28 CRON[2764]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:39:01 ip-172-31-35-28 CRON[2765]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:01 ip-172-31-35-28 CRON[2764]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:38 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cur
/main/linper.sh
Mar  6 06:39:38 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=100
Mar  6 06:39:39 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:40:01 ip-172-31-35-28 CRON[2783]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
Mar  6 06:40:01 ip-172-31-35-28 CRON[2784]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=
```

https://github.com/montysecurity/linper

README

linper

linux persistence toolkit

files

- countermeasures/ - ways to harden/monitor common persistence mechanisms
- powershell/ - placeholder for the eventual winper
- CONTRIBUTE.md - notes on contributing to this project
- README.md - or not
- TODO.md - planned fixes & enhancements
- linper.sh - execute me

features

- enumerate programs that can be used to execute a reverse shells and ways to make them persist a reboot
- automatically install reverse shells with all the required syntax, redirection, and pipes to minimize printing errors to screen or interrupting normal functions and processes
- supply custom crontab schedules for reverse shells
- look through /etc/shadow for accounts that can login via a password
- support for a stealth mode and the ability to clean up after itself
- place a function in ~/.bashrc to intercept and exfil sudo passwords
- place php reverse shells in web server directories